Linked List INSERTION

DAY 37
16/08/25

Inserting a Node at a given position in a Linked List

Case ① if (Insertpos <= 0)
its a Invalid position for Insertion

Case ② if (Insertpos == 1)
Insert at Begining of linked List

Case ③ if (Insertpos = size+1)
Insert at End of Linked List

Case ④ if (Insertpos > size+1)
Invalid Insert position

🔷 **Steps for insertAtPosition(int position, int data)**

1. Check position validity

- If `position <= 0` → invalid (print `"Invalid Position"`).

2. Insert at beginning

- If `position == 1` → call `insertAtBegining(data)`.
- Example: inserting at pos = 1 in `10 → 20 → 30` gives `5 → 10 → 20 → 30`.

3. Insert at end

- If `position == size + 1` → call `insertAtEnd(data)`.
- Example: inserting at pos = 4 in `10 → 20 → 30` gives `10 → 20 → 30 → 40`.

4. Traverse to (position - 1)th node

- Start from head with `curPos = 1`.
- Move forward until `curPos < position - 1`.
- After loop ends, `cur` will point to the **previous node** where insertion should happen.
- Example: Insert 25 at pos = 3 in `10 → 20 → 30 → 40`.
  - Traverse until `curPos = 2`.
  - Now `cur` points to `20`.

5. Check if current node is null

- If `cur == null`, position is invalid → print `"Invalid Position"`.

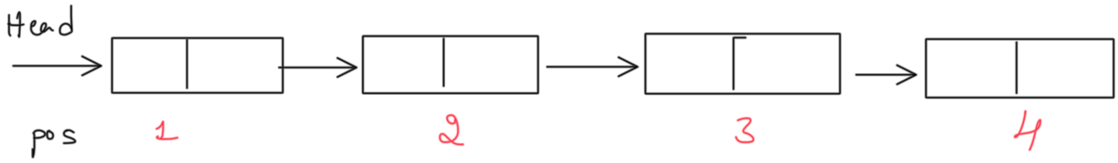
6. Create and link new node

- Create `newNode = new Node(data)`.
- Adjust pointers:
  1. `newNode.next = cur.next;`
  2. `cur.next = newNode;`

7. Increase size

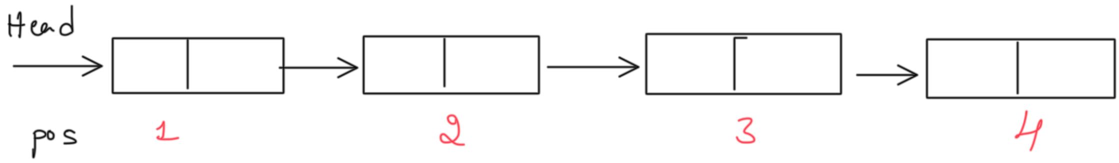
- `size++` since new node added.

Consider a linked list

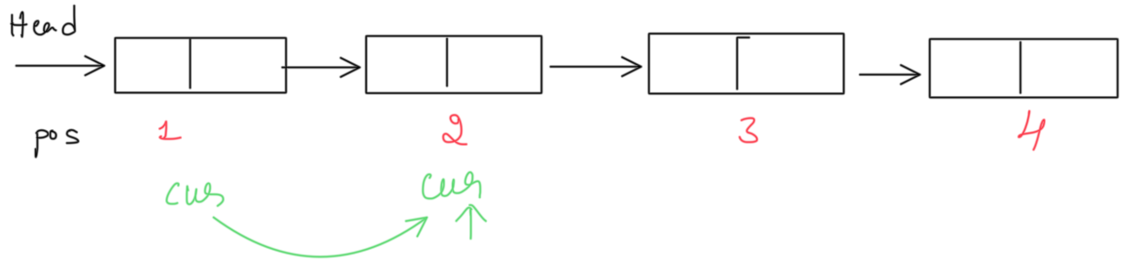


Insert At position 3 pos = 3

take currNode as Head

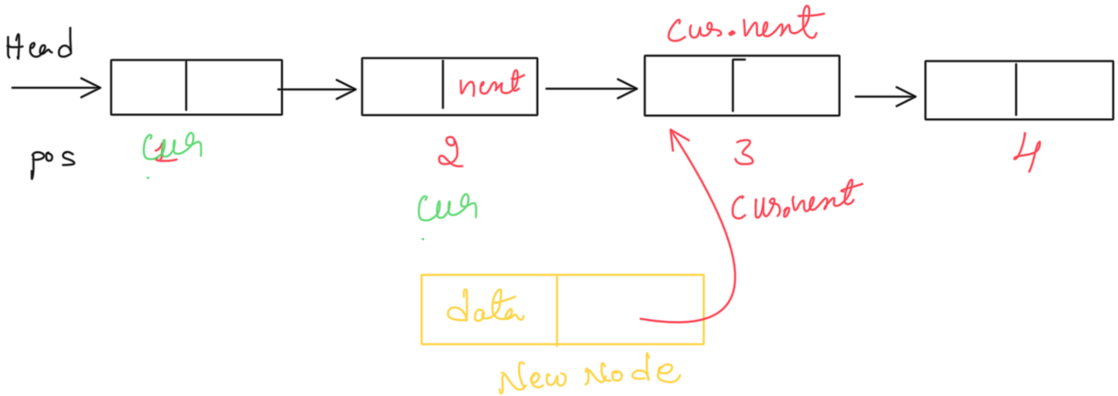


move until position 3 - 1 = 2

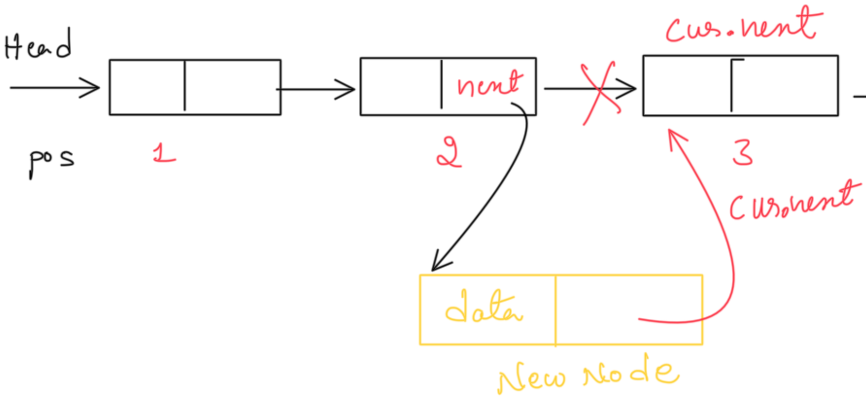


Create a NewNode with data → Insert data

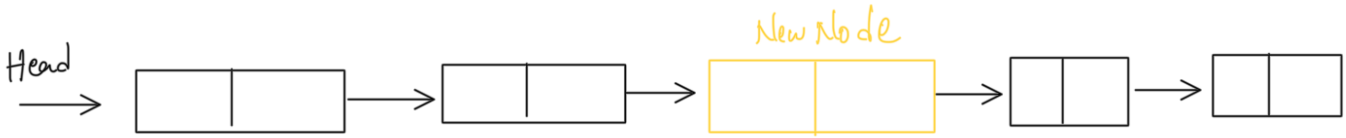
Step 1 : NewNode.next = curr.next



step 2 update curr.next to New Node



Linked list After successfull Insertion



EXPLORER ···

DSA

LinkedList

- DSA_Application.java 1
- Node.java 1
- SinglyLinkedList.java 1

SinglyLinkedList.java 1 × | Extension: ChatGPT – Work with Code on macOS | DSA_Application.java 1 | Node.java 1

LinkedList > SinglyLinkedList.java > Java Language Support > SinglyLinkedList

```java
public class SinglyLinkedList {
    }

    public void insertAtPosition(int position,int data){

        if(position<=0){
            System.out.println(x:"Invalid Position ");
            return;
        }

        if(position == 1) {
            insertAtBegining(data);
            return;
        }

        if (position == size + 1) {
            insertAtEnd(data);
            return;
        }

        Node cur = head;
        int curPos = 1 ;

        while(curPos < position-1 && cur!=null){
            cur = cur.next;
            curPos++;
        }//loop to travel prev node of insertposition

        if(cur == null) {
            System.out.println(x:"Invalid Position");
            return;
        }

        Node newNode = new Node(data);

        newNode.next = cur.next;
        cur.next = newNode;

        size++;

    }
```

TIMELINE

OUTLINE

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JAVA PROJECTS

EXPLORER ···

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Node.java 1
SinglyLinkedList.java 1

SinglyLinkedList.java 1 × | Extension: ChatGPT – Work with Code on macOS | DSA_Application.java 1

LinkedList > SinglyLinkedList.java > Java Language Support > SinglyLinkedList > testAllOperations

```java
public class SinglyLinkedList {
    public void insertAtPosition(int position,int data){

    }

    public void testAllOperations(){

        this.insertAtBegining(data:3);
        this.insertAtBegining(data:2);
        this.insertAtBegining(data:1);

        this.printAllNodesData();

        System.out.println("Size "+size);

        this.insertAtEnd(data:4);
        this.insertAtEnd(data:5);
        System.out.println("Size "+size);

        this.insertAtEnd(data:6);
        this.printAllNodesData();
        System.out.println("Size "+size);

        this.insertAtPosition(position:4, data:20);
        this.printAllNodesData();

        this.insertAtPosition(position:1, -1);
        this.insertAtPosition(position:0, data:2);
        this.printAllNodesData();

        this.insertAtPosition(position:20, data:999);
        this.insertAtPosition(size, data:12 );
        this.printAllNodesData();
        System.out.println("Size "+size);

        this.insertAtPosition(size+1, data:55 );
        this.printAllNodesData();
        System.out.println("Size "+size);


    }

}
```

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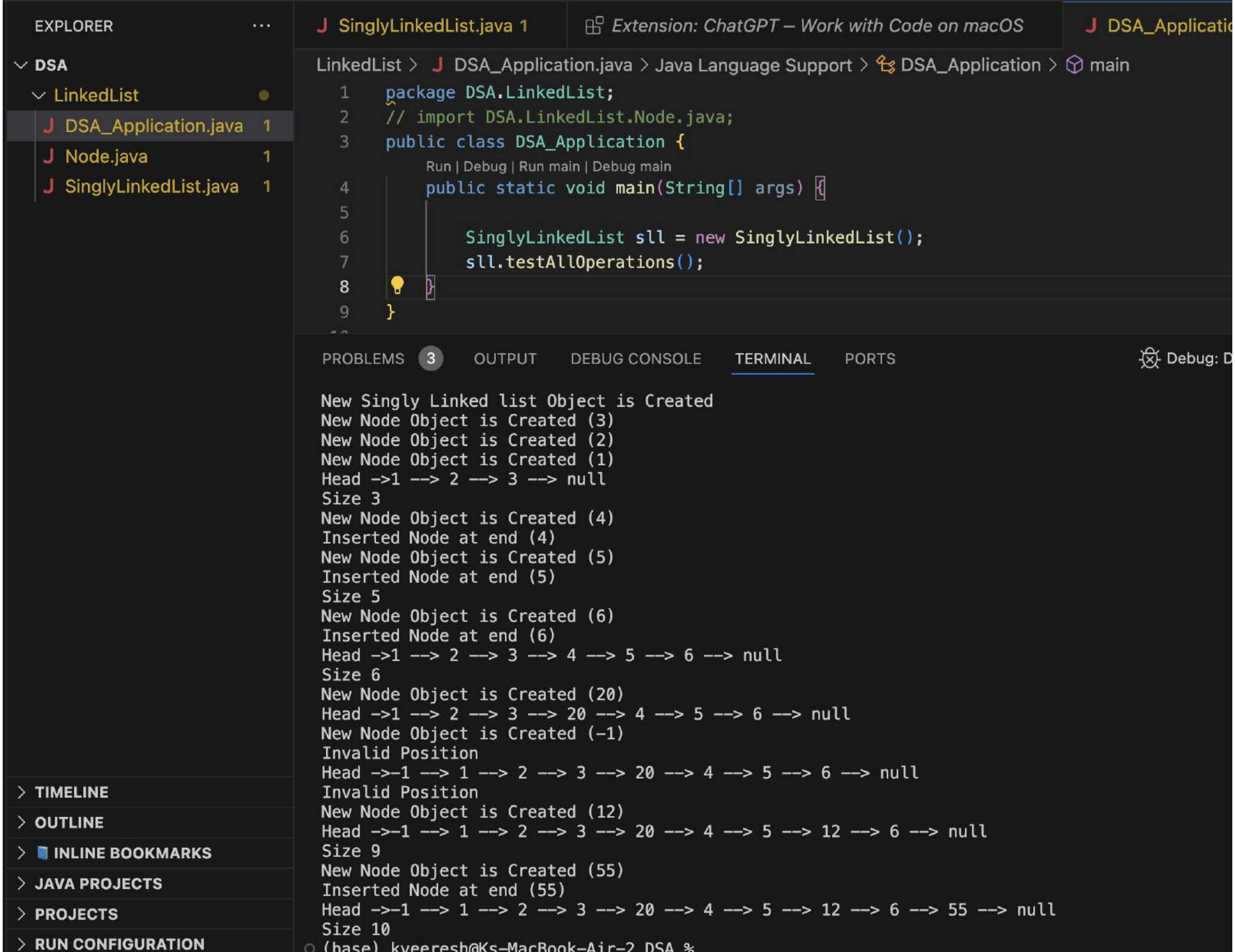
RUN CONFIGURATION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PROJECTS
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SinglyLinkedList.java 1
Extension: ChatGPT – Work with Code on macOS
DSA_Applicati
LinkedList > DSA_Application.java > Java Language Support > DSA_Application > main
package DSA.LinkedList;
// import DSA.LinkedList.Node.java;
public class DSA_Application {
Run | Debug | Run main | Debug main
public static void main(String[] args) {
SinglyLinkedList sll = new SinglyLinkedList();
sll.testAllOperations();
}
}
PROBLEMS 3
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
Debug: D
New Singly Linked list Object is Created
New Node Object is Created (3)
New Node Object is Created (2)
New Node Object is Created (1)
Head ->1 --> 2 --> 3 --> null
Size 3
New Node Object is Created (4)
Inserted Node at end (4)
New Node Object is Created (5)
Inserted Node at end (5)
Size 5
New Node Object is Created (6)
Inserted Node at end (6)
Head ->1 --> 2 --> 3 --> 4 --> 5 --> 6 --> null
Size 6
New Node Object is Created (20)
Head ->1 --> 2 --> 3 --> 20 --> 4 --> 5 --> 6 --> null
New Node Object is Created (-1)
Invalid Position
Head ->-1 --> 1 --> 2 --> 3 --> 20 --> 4 --> 5 --> 6 --> null
Invalid Position
New Node Object is Created (12)
Head ->-1 --> 1 --> 2 --> 3 --> 20 --> 4 --> 5 --> 12 --> 6 --> null
Size 9
New Node Object is Created (55)
Inserted Node at end (55)
Head ->-1 --> 1 --> 2 --> 3 --> 20 --> 4 --> 5 --> 12 --> 6 --> 55 --> null
Size 10
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